

Innovation on technological “islands”: domain contrast, boundary spanning, knowledge depth and breadth

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Abstract

Prior literature has long examined innovation as a recombination process within or across the boundaries of technological domains. However, limited attention is paid to boundaries per se. Building upon recent development of categorical contrast, this study distinguishes domains with crisp boundaries from those with fuzzy boundaries and examines their effects on innovation outputs. Analyzing a large sample of US patents, we find that spanning *crisp* boundaries is more likely to generate impactful inventions but at the same time leads to significantly higher recombinant uncertainty. We continue to explore what types of inventors are better able to span such types of domain boundaries. Focusing specifically on the role of inventors' knowledge expertise, we find that while both knowledge depth and breadth enhance the impact of technologies that span crisp boundaries, knowledge breadth is also found to escalate the associated uncertainty. Our emphasis on the contrast of technological domains contributes to the literature on recombinative innovation and boundary spanning.

JEL classification: O32, O34

1. Introduction

Innovation usually relies on the recombination of technological components (Schumpeter, 1934; Fleming, 2001). The conditions for recombining components exist as technological space is not a seamless web, but fragmented into many domains (Carnabuci and Bruggeman, 2009; Tell, 2017). Technologies are densely connected *within* domains but loosely connected *across* domains (Hargadon, 2002). In their innovation process, inventors may either utilize components within the same domain or span boundaries to combine technologies across different domains (Gruber *et al.*, 2013). Oftentimes, recombination across technological boundaries is more likely to generate impactful innovations than that in a single domain (Hargadon, 1998; Rosenkopf and Nerkar, 2001; Fleming *et al.*, 2007; Lo and Kennedy, 2015). As such, a large amount of work in the innovation literature has examined the antecedents and consequences of boundary-spanning recombination (Nemet and Johnson, 2012; Gruber *et al.*, 2013; Kaplan and Vakili, 2015; Ferguson and Carnabuci, 2017). Despite extensive research on cross-boundary innovation, however, prior studies pay much less attention to boundaries per se.

This paper emphasizes the heterogeneity of technological boundaries and highlights that the same degree of boundary spanning will lead to very different innovation outputs when the boundaries being spanned differ. More specifically, building upon organizational research of categorical contrast (Hannan *et al.*, 2007), we underscore that technological domains differentiate from each other in terms of their *contrast*. Contrast reflects the extent to which a domain has a clear-cut

boundary that distinguishes it from the other domains (Kovács *et al.*, 2021). Therefore, high-contrast domains (i.e., crisp domains) have sharp and distinctive boundaries, whereas domains lacking contrast (i.e., fuzzy domains) are characterized by blurry and permeable boundaries (Hannan, 2010; Carnabuci *et al.*, 2015). Technological domains do show substantial heterogeneity in terms of contrast. In our empirical context of the, US Patent and Trademark Office (USPTO) system, for instance, technology class 257 (active solid-state devices) is crisp because most innovations within the class are highly typical with distinctive features; class 435 (chemistry: molecular biology and microbiology) has fuzzy and permeable boundaries because many of its innovations are produced with dispersed components outside the class, lacking internal coherence or distinctiveness. Focusing on the contrast of technological boundaries, our aims are twofold.

First, we examine whether and how contrast influences the effect of boundary spanning on innovation outcomes. The innovation literature has firmly established the importance of spanning boundaries, as impactful innovations are usually the outcomes of bridging disparate technological domains (Rosenkopf and Nerkar, 2001; Yayavaram and Ahuja, 2008; Guan and Yan, 2016). In analyzing boundary-spanning innovation, however, scholars have often overlooked the heterogeneity of those boundaries. In other words, most studies have implicitly assumed that technological boundaries are homogenous so that only the *degree* of boundary spanning matters for innovation outputs. Extending this stream of research, we distinguish technological domains with crisp boundaries from those with fuzzy boundaries (Carnabuci *et al.*, 2015; Kovács *et al.*, 2021) and employ the knowledge recombination perspective (Xiao *et al.*, 2022) to analyze the implications of boundary-spanning innovation in different domains. On the one hand, we conjecture that spanning crisp boundaries is more likely to generate impactful innovations because technologies in crisp domains are clearly distinctive from the external world, thus increasing the possibility of creating novel recombination (Fleming *et al.*, 2007; Uzzi *et al.*, 2013; Aharonson and Schilling, 2016). On the other hand, however, it may also escalate recombinant uncertainty because the distinctive nature of crisp domains means that there exists little prior knowledge about the potential of combinational relationships that span crisp boundaries (Fleming, 2001).

Second, and more importantly, we explore further what types of inventors are better able to span crisp boundaries. Whereas prior research has suggested that boundary-spanning activities may suffer deficiencies in the production of innovations (Hsu *et al.*, 2009; Kaplan and Vakili, 2015), less is known about how inventors may improve their efficiencies in those activities. More concretely, if spanning crisp boundaries is a double-edged sword that increases both inventive impact and uncertainty, it is interesting to understand what may help inventors to reap the associated benefits with less uncertainty. We focus specifically on inventors' knowledge depth and breadth, both of which are important for their inventive outcomes (Boh *et al.*, 2014; Mannucci and Yong, 2018). Building upon the knowledge combination view (Xiao *et al.*, 2022), we argue that while knowledge depth in a particular domain is crucial for inventors to span crisp boundaries, knowledge breadth may breed more uncertainty in the process.

We test our hypotheses using a comprehensive sample of patents approved in the United States between 2001 and 2010. The context is appropriate for this study because of its well-structured technological space, i.e., the USPTO classification system (Fleming, 2001; Hall *et al.*, 2001; Nemet and Johnson, 2012). Specifically, the system categorizes each patent into an original technology class based on its most important knowledge contribution (Fleming, 2001; Carnabuci and Bruggeman, 2009). We can thus easily identify the primary technological domains for all patents (Carnabuci *et al.*, 2015). Patents are also frequently assigned to subclasses from other technology classes that provide supporting components, which allows us to investigate boundary-spanning recombination (Fleming, 2001; Gruber *et al.*, 2013; Lo and Kennedy, 2015; Ferguson and Carnabuci, 2017). The empirical results lend general support to most of our hypotheses.

2. Theory and hypotheses

2.1 Recombinative innovation and boundary spanning

The ultimate source of creativity and innovation is recombination (Schumpeter, 1939; Fleming, 2001). In the innovation process, inventors search for optimal solutions to their problems, by either recombining existing technological (and/or non-technological) components in a new way or introducing novel components (Keupp and Gassmann, 2013; Guan and Yan, 2016). Components, or “factors” in Schumpeter’s (1939) terms, are the fundamental building blocks of an innovation (Rosenkopf and Nerkar, 2001), with each of them as “an embodiment of a core concept based on a distinct scientific or engineering principle” (Xiao *et al.*, 2022: 1728). Take a new automobile as an example of innovation—components would include its frame, power source, brake system, and so on (Fleming, 2001). Adopting this Schumpeterian recombination perspective, Fleming (2001) provides the first empirical test on recombining components. He shows solid evidence that whereas new combinations lead to less impactful inventions on average, it also implies an increase in recombinant uncertainty that can result in both breakthrough and failure.

Building upon Fleming’s (2001) seminal work, extensive innovation studies have investigated various forms of recombination. Scholars have paid substantial attention to the process of boundary-spanning recombination (Rosenkopf and Nerkar, 2001; Carnabuci and Operti, 2013; Ferguson and Carnabuci, 2017). In the technological space, components are not randomly located but organized by domain-specific knowledge boundaries that distinguish them according to features, characteristics, and/or canonical principles (Nemet and Johnson, 2012; Tell, 2017). Components within the same domain boundaries thus exhibit commonality and are more densely connected with each other; components from different domains vary more substantially in terms of key functional and/or structural features (Hargadon, 2002; Carnabuci and Bruggeman, 2009). For instance, components in the USPTO class 257 concentrate on active solid-state devices (e.g., transistors and solid-state diodes), whereas components from class 435 are mostly related to molecular biology and microbiology.¹

Recombination of components can thus occur within a single domain boundary or across two or more domain boundaries. It has become a stylized fact that innovations are more impactful when they combine components across domain boundaries than when they draw from a single domain. For instance, Rosenkopf and Nerkar (2001) show that in the global optical disk industry, inventions spanning boundaries have a higher overall impact as indicated by forward citations, albeit a lower domestic impact. Ferguson and Carnabuci (2017) find similar evidence that the diversity of an invention’s prior arts is positively related to its technological impact. Extending the linear relationship between boundary spanning and technological impact, Lo and Kennedy (2015) show that in the nanotechnology area, while a patent’s forward citations increase with the number of technology classes that it builds on, this effect diminishes as the number of classes continues to grow.

2.2 Contrast of technological domains

While prior literature has focused on the *degree* of boundary spanning in the innovation process (Rosenkopf and Nerkar, 2001; Ferguson and Carnabuci, 2017), we direct attention to the heterogeneity of domain boundaries. Building upon category research (Hannan *et al.*, 2007; Hsu *et al.*, 2009), we emphasize that the boundaries of technological domains differ along an important dimension—categorical contrast. Contrast indicates the extent to which a domain can be distinctively identified from others (Kovács and Hannan, 2010; Carnabuci *et al.*, 2015; Kovács *et al.*, 2021). High-contrast or crisp domains are featured with clear-cut and sharp boundaries that distinguish them from other domains. They may be conceptually thought of as “islands” in the technological landscape that are isolated from the other space. Conversely, low-contrast or fuzzy domains have porous and fuzzy boundaries, being more blended with other domains. Earlier research on categorical contrast concentrates mostly on cultural products (Hsu *et al.*, 2009).

¹ While we focus on domain-specific boundaries, knowledge boundaries can also be conceptualized along other dimensions such as individual boundaries, task-oriented boundaries, and spatial and temporal boundaries. See Tell (2017) for more details.

Analyzing the food industry, for instance, Kovács and Hannan (2010) suggest that restaurants are more appealing to market audiences if they operate in a crisp food category that shows strong distinctiveness relative to others. However, spanning crisp boundaries in the cultural fields may also undermine product appeal more than does spanning fuzzy boundaries. That is because crisp cultural categories possess stronger cultural codes, and spanning them may result in stronger code clashes.

Recent studies have started to highlight that the process of technological innovation and scientific discovery is also subject to categorical contrast (Kovács *et al.*, 2021). In particular, Carnabuci *et al.* (2015) show that semiconductor firms choose innovation directions according to the contrast of technological domains. Similarly, Pontikes and Barnett (2015) report that software start-ups make decisions about product entry and exit based on the contrast of market categories. Kovács *et al.*'s (2021) recent paper finds that inventions in crisp technological domains garner more attention and accrue more citations than inventions in fuzzy domains. Extending research in this stream, we aim to understand whether and how domain contrast may influence *boundary-spanning* technological innovation.²

First, we argue that spanning crisp boundaries are more likely to generate impactful innovations than spanning fuzzy boundaries. The knowledge recombination view posits that impactful innovations that exert strong influences on future technological developments (e.g., as commonly reflected by forward citations) are usually the results of recombining components with divergent functions or attributes (Hargadon, 2002; Xiao *et al.*, 2022). Being constrained by its clear-cut boundary, components in a crisp domain exhibit both a high degree of *internal* coherence among themselves (Carnabuci *et al.*, 2015) and a high degree of distinctiveness from *external* components in other domains (Hannan *et al.*, 2007; Hsu *et al.*, 2009; Kovács *et al.*, 2021). As such, spanning crisp boundaries to import external components will increase the possibility of developing novel innovations, as this may yield functions and meanings that are above and beyond that of existing ones (Keupp and Gassmann, 2013; Aharonson and Schilling, 2016). Ultimately, the enhanced novelty will lead to innovations that are more useful and impactful (Fleming *et al.*, 2007; Uzzi *et al.*, 2013; Kaplan and Vakili, 2015).

By contrast, spanning fuzzy domains may not bring the same amount of benefits. Components in a fuzzy domain exhibit less coherent or distinctive features relative to other domains (Carnabuci *et al.*, 2015). Rather, they show greater ambiguity and generality because the porous boundaries allow a wide range of diverse components to blend in. While internal incoherence allows inventors to generate flexible combinations within a fuzzy domain (Pontikes and Barnett, 2015), lacking distinctiveness also limits the possibility of developing novel combinations when spanning the fuzzy boundaries. As a result, boundary spanning will contribute less value to the process of innovation when a domain is fuzzy than when it is crisp. We predict that,

H1A: Inventions that span crisper boundaries are more impactful than inventions that span fuzzier boundaries.

However, whereas spanning crisp boundaries helps generate impactful innovations, it may also escalate recombinant uncertainty. Recombinant uncertainty refers to variance in innovation outcomes (Fleming and Sorenson, 2001). While uncertainty often results in a higher chance of inventive failures, it is also an important source of significant advances and occasional breakthroughs (Taylor and Greve, 2006). It is well established that recombinant uncertainty increases with inventors' unfamiliarity of combinational relationships (Fleming, 2001). On the one hand, prior usage of certain combinations enables inventors to gain knowledge and familiarity about them. As inventors learn which combinations are less useful, inventors can avoid those

² It is important to note that the concept of domain contrast is related to, but different from, several important concepts in prior literature (e.g., technological diversity, distance, familiarity, and generality). Contrast is a *domain-level* factor, indicating the extent to which a domain has a clear-cut boundary that distinguishes it from the others. Familiarity is a *component (or combination)-level* factor, referring to the extent to which a component has been used in prior inventions. Technology distance and diversity are *invention-level* measures, indicating the extent to which the components used in an invention are different from one another and the extent to which the components used in an invention are from different domains, respectively. Technology generality is also an *invention-level* factor, reflecting how dispersed an invention's forward citations are distributed.

approaches, thereby reducing the risk of generating failures (Levinthal and March, 1993). On the other hand, however, familiarity may also drive inventors toward local search and exploitative innovation (Rosenkopf and Nerkar, 2001), which undermines the possibility of achieving technological breakthroughs (Xiao *et al.*, 2022). Either way, familiarity of combinational relationships leads to reduced uncertainty.

Being isolated by sharp boundaries, components in a crisp domain are, by definition, less frequently combined with external components (Carnabuci *et al.*, 2015). There are fewer pre-existing innovations that span crisp boundaries, from which inventors can learn. The inability to draw on prior knowledge increases uncertainty in the innovation process, as inventors step into the unknown in search of a desirable combination, leading to greater variance in innovation outcomes (Fleming, 2001). Moreover, the knowledge recombination process is not frictionless. Rather, it requires translating and adapting knowledge inputs from other domains into a focal domain (Gittelman and Kogut, 2003; Tell, 2017). This process is particularly unpredictable when a focal domain has distinctive features and epistemology within crisp boundaries. By contrast, spanning fuzzy boundaries is less cognitively challenging (Pontikes and Barnett, 2015). There may also exist more prior knowledge with regard to their combinational potentials and paths because components in a fuzzy domain are less separated from external components. Inventors can therefore learn from preexisting efforts and limit the possibility of wild and unexpected outcomes. As such, we expect:

H1B: Inventions that span crisper boundaries are associated with higher uncertainty than inventions that span fuzzier boundaries.

2.3 Knowledge depth and breadth for spanning crisp boundaries

After theorizing the outcomes of spanning crisp boundaries, we continue to explore what types of inventors are better able to span those boundaries to take advantage of the associated novelty and at the same time mitigate the uncertainty. Inventors differ in education, knowledge, and experience, leading to very different boundary-spanning behaviors and outcomes. Focusing on education background, for instance, Gruber *et al.* (2013) show that inventors with a scientific education are more likely to span technological boundaries than inventors with an engineering degree because scientific education provides them with broader and more abstract knowledge. Drawing attention to inventors' social structures, Fleming *et al.* (2007) show that brokerage positions enhance the creation of new combinations, which is further moderated by inventors' experiences and extended networks. Emphasizing organizational contexts, Rosenkopf and Almeida (2003) suggest that strategic alliances and employee mobility can serve as bridges to distant knowledge and, thus, enable organizations to overcome the constraints of localized search.

To understand who can better span crisp boundaries, we focus on two important aspects of inventors' knowledge expertise: depth and breadth (Mannucci and Yong, 2018). Knowledge depth refers to the level of domain-specific knowledge that inventors have accumulated in the focal domain of a particular innovation (Boh *et al.*, 2014; Wang *et al.*, 2017). It enables inventors to understand fundamental components and the complex relationships between them so that they can generate impactful inventions (Katila and Ahuja, 2002). Because knowledge depth is domain-specific, the same inventor may have different levels of knowledge depth in different technological domains, which may also have different degrees of categorical contrast. Knowledge breadth, on the other hand, refers to the diversity in knowledge, skills, and experience that inventors hold (Taylor and Greve, 2006).³ Knowledge breadth arms inventors with variations and flexibility so that they are able to loosen up existing relationships and create new combinations (Mannucci and Yong, 2018), particularly when inventors are constrained by certain social structures (Fleming

³ While some studies conceptualize depth and breadth as two ends of the same dimension (e.g., Carnabuci and Operti, 2013; Melero and Palomeras, 2015), emphasizing the trade-offs between focusing on specialized areas and broadening expertise domains, we follow the other perspective (e.g., Boh *et al.*, 2014; Caner and Tyler, 2015; Mannucci and Yong, 2018) to treat them as two dimensions. That is, we see that there exist both domain experts with knowledge of many other domains (i.e., high in both dimensions) and novices with a narrow focus (i.e., low in both).

et al., 2007; Carnabuci and Operti, 2013). While prior research has underscored the importance of the two factors in the generic innovation process, we focus on their specific roles in the process of spanning crisp boundaries.

2.3.1 Knowledge depth

We argue that knowledge depth is important for inventors to span crisp boundaries. Deep knowledge in a domain means an in-depth understanding of its components, relationships, assumptions, and principles (Xiao *et al.*, 2022). As an inventor accumulates experience from experimenting with various components in a given domain, he or she will be better able to understand their core attributes, as well as potential problems when combining them in different ways (Ericsson and Lehmann, 1996; Boh *et al.*, 2014). This will boost inventors' ability to identify valuable relationships, such that they can utilize knowledge in many different ways that are not apparent to novice inventors (Katila and Ahuja, 2002). In other words, it is difficult for inventors to see new sources of value without understanding the basic assumptions behind the existing sources (Kaplan and Vakili, 2015), and these insights can only result from a deep appreciation of technological domains (Weisberg, 1999).

It is particularly so when they innovate in crisp domains. Crisp domains have more coherent internal features, distinctive epistemic communities, and even unique criteria for assessing novelty, originality, and value (Kovács and Hannan, 2010). Knowledge depth thus becomes even more important for inventors to gain an understanding of distinctive features in crisp domains and identify "what rules to break" (Taylor and Greve, 2006). Deep knowledge, in other words, is a crucial prerequisite if inventors are to successfully import external knowledge into crisp domains. Novice inventors without sufficient expertise will be largely disadvantaged in the process, as they are unable to make relevant inferences about what it means and how it works in a crisp domain, which are not obvious from the surface (Chi *et al.*, 1989; Boh *et al.*, 2014). When spanning a crisp domain, therefore, lacking domain-specific knowledge will likely lead to poorer innovation outcomes and higher rates of failure, albeit it may occasionally lead to an out-of-the-box hit. By contrast, fuzzy domains present greater flexibility that allows novice inventors with limited domain-specific knowledge to easily fit in (Pontikes and Barnett, 2015) so that the advantage of knowledge depth may be less pronounced when spanning fuzzy boundaries.

H2A: Knowledge depth enhances the impact of inventions that span crisper boundaries, as compared to those spanning fuzzier boundaries.

H2B: Knowledge depth decreases the uncertainty of inventions that span crisper boundaries, as compared to those spanning fuzzier boundaries.

2.3.2 Knowledge breadth

Knowledge breadth across various domains is known as important for recombinative innovation because it improves the possibility that inventors generate new combinations of disparate components (Katila and Ahuja, 2002; Fleming *et al.*, 2007; Carnabuci and Operti, 2013; Gruber *et al.*, 2013). However, it does not necessarily mean that their recombinations are more useful or impactful (Taylor and Greve, 2006). Indeed, Boh *et al.*'s (2014) study in 3M's innovation show that knowledge breadth has no significant effect on technological impact. Similar evidence is found in Fleming *et al.*'s (2007) large-scale study that knowledge breadth does not enhance the impact of inventions, unless inventors have established certain types of networks. This is possible because although knowledge breadth enlarges the pool of diverse components, it also leads to cognitive overload and a lack of specialty that may dampen the innovation process (Taylor and Greve, 2006; Mannucci and Yong, 2018).

We contend that the value of knowledge breadth is also contingent on domain contrast, such that broad knowledge scope may be important for spanning fuzzy boundaries. As stated earlier, components in a fuzzy domain are less coherent, and its epistemic communities are accommodating more diverse traditions, approaches, and principles (Pontikes, 2012; Carnabuci *et al.*, 2015). Inventors with broad knowledge are more advantaged in navigating such domains because innovating in fuzzy domains requires cognitive schemas and cause-effect understandings that are much

more flexible and diverse (Dougherty, 1992; Carnabuci and Operti, 2013). As such, a high-level of knowledge breadth is useful to support inventors in generating innovations that span fuzzy boundaries.

By contrast, when spanning crisp boundaries, knowledge breadth may backfire. First, components in crisp domains are coherent and consistent, which reduces the demand for a very broad scope and cognitive schemas in spanning such domains. Second, too wide scope may hinder inventors from developing more focused insights into crisp domains because of the induced cognitive overload and dispersed attention (Hannan *et al.*, 2007; Kaplan and Vakili, 2015; Mannucci and Yong, 2018). However, since a more focused search is crucial for spanning crisp domains, inventors with a broad knowledge scope may face more uncertainty and produce superficial or incremental work. Taken together, we expect:

H3A: Knowledge breadth decreases the impact of inventions that span crisper boundaries, as compared to those spanning fuzzier boundaries.

H3B: Knowledge breadth increases the uncertainty of inventions that span crisper boundaries, as compared to those spanning fuzzier boundaries.

In sum, we conjecture that while knowledge depth is more important for spanning crisp domains, knowledge breadth is more beneficial for innovations that span fuzzy domains. Take the software market as an example, where the domains of “enterprise software” and “electronic design automation” are plausibly fuzzy and crisp, respectively (Pontikes, 2012). Our theory would suggest that knowledge breadth may help inventors import knowledge into developing enterprise software because this domain is internally less coherent, supporting inventors with diverse knowledge scope to fit it. Knowledge depth, by contrast, is important for inventors to import knowledge to the crisp domain of electronic design automation that has distinctive design principles and epistemic communities. Consider also academic disciplines, where economics and management are arguably crisp and fuzzy, respectively. While it is essentially impossible to bring management knowledge into economics unless scholars have deep knowledge of economics (e.g., its theory, languages, methods, and debates), the opposite is not true. Instead, scholars with broad knowledge in other fields (e.g., econometrics) can more easily contribute to management, even without deep knowledge about management.⁴

3. Empirics

We test our hypotheses by analyzing all of US patents between 2001 and 2010. Patenting is a context commonly used for analyzing recombinative and boundary-spanning innovations because patents are assigned to technology classes that can be conceptualized as domains in the technological space (Fleming, 2001; Nemet and Johnson, 2012; Lo and Kennedy, 2015; Kovács *et al.*, 2021). We collect data mainly from PatentsView supported by the USPTO, which provides convenient retrieval of patent information (Bruce *et al.*, 2019; Righi and Simcoe, 2019). We also supplement the PatentsView with the National Bureau of Economic Research Patent Database that provides cleaner information on patent assignees (Hall *et al.*, 2001). We cover patents in all industries, which enables us to make a more comprehensive coverage of technology classes (Nemet and Johnson, 2012). While the PatentsView database includes patents that were granted till 2015, we choose to analyze patents up to 2010 because forward citations for patents after 2010 would be less informative of their impact.

3.1 Dependent variable

3.1.1 Impact and uncertainty of patents

The number of forward citations is widely used to measure the technological impact of patents (Nemet and Johnson, 2012; Carnabuci *et al.*, 2015). Although forward citations may be linked

⁴ We thank one reviewer for suggesting this vivid example.

to other factors or may not fully capture the impact on future inventions (Nelson, 2009), it gives a good albeit imperfect indication of how a patent shapes technological development, and for this reason, it is found to be highly correlated with technological excellence and market value (Hall *et al.*, 2005; Singh and Fleming, 2010; Kaplan and Vakili, 2015).

While the total number of forward citations was commonly used for measuring patent impact, recent studies emphasize that citations are often added by patent examiners (Righi and Simcoe, 2019). It is hence important to remove such types of forward citations to more accurately reflect a patent's direct impact on technological communities. To measure the technological impact, we collect data on the sources of citations, remove examiner citations, and count the number of forward citations for each patent that are not added by examiners up to 2019. In regressions, we use the log transformation of forward citations to reduce skewness. One is added before the transformation to avoid taking the natural log of a zero value.

To measure the recombinant uncertainty of patents, we use their conditional variance of forward citations (Fleming and Sorenson, 2004; Ganco, 2017). Specifically, we first run fixed-effects regressions (as explained later) to estimate the mean of forward citations, based on which we generate a predicted number of forward citations for each patent and their predicted residuals. We then use the natural logarithm of squared residuals to indicate the conditional variance or recombinant uncertainty of patents (Ganco, 2017).

3.2 Key explanatory variables

3.2.1 Contrast of tech domain

We use information on patents' class assignments to measure the contrast of technology classes, by leveraging the hierarchical structure of USPTO classes and subclasses and treating each class as a distinct technological domain (Carnabuci *et al.*, 2015). More concretely, based on "claims" in application documents, each patent is assigned to one original main class and one or multiple subclasses within or outside the original class (Fleming, 2001).⁵ The original class of a patent encompasses its primary inventive function, whereas subclass assignments outside the original class capture complementary technologies that are useful to the operation of the patented technology. We hence refer to a patent's original class as its primary technological domain.

Following prior studies (Carnabuci *et al.*, 2015; Kovács *et al.*, 2021), we first calculate the proportion of a patent's subclasses that belong to its original class. A larger proportion indicates that the patent is a typical exemplar of its original class. Conversely, when most subclasses fall outside a patent's original class, the patent represents the class poorly. To calculate the contrast of a class, we then average the proportions for all patents in the class during the last 5 years ($t - 4$ to t). As such, each class has a contrast ranging between 0 and 1, which is time-variant. A large value indicates a crisp domain as most patents show a clear membership to the original class. Figure 1 shows the contrast of five selected technology classes by years, according to our measure. While the degree of contrast is stable for some classes, others have a more turbulent contrast over time (e.g., class 438—semiconductor device manufacturing: process).⁶

3.2.2 Domain spanning

To capture the extent to which a patent spans its primary domain (i.e., original class), we create two measures. First, we use a dichotomous variable, *domain spanning (dummy)*, with a value of 1 if at least one subclass of the patent belongs to technology classes that are outside its original class, and 0 otherwise. For instance, patent US4467150 (electronic keyboard) is classified in original class 200, but three out of its five subclass assignments (361/398, 400/479, and 400/488) belong

⁵ PatentsView provides both the original and current class assignments for each patent. The USPTO classifications scheme has been amended and revised over the years, as classes are sometimes retroactively consolidated and/or discontinued. In our study, we apply the original classification assignment as we focus on technological recombinations as they occur.

⁶ Although the dynamics of domain contrast are not the focus of this study, it is interesting to adopt an evolutionary perspective (Carnabuci and Bruggeman, 2009) to examine how and why technology classes differ in terms of the growth or decline of their contrast. In other words, what drives an island in the technology space to become more connected over time, or the other way around? In an additional analysis not reported here, we try to control for the dynamics, by adding a new variable, *contrast change*. It is measured as domain contrast at year t minus domain contrast at $t - 1$ divided by domain contrast at $t - 1$. The results remain consistent.

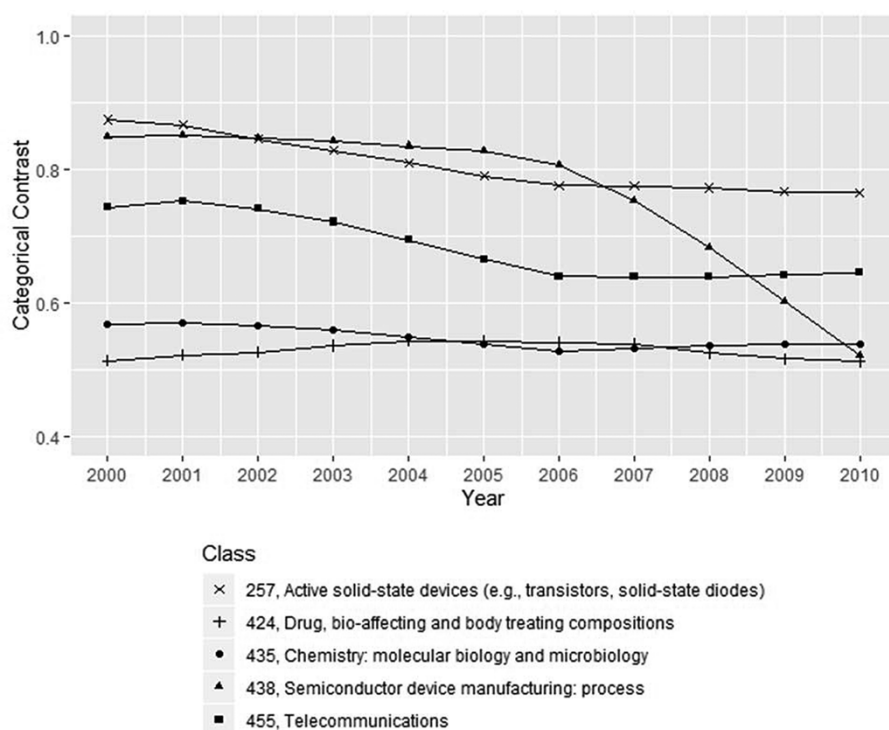


Figure 1. Contrast for selected patent classes

to other technology classes than class 200. It is hence assigned with a value of 1 for *domain spanning (dummy)*. Second, we create a continuous variable, *domain spanning (ratio)*, measured as the number of a patent’s subclasses belonging to other classes divided by the total number of its subclasses. For patent US4467150, the measure is accordingly 0.6 ($=3/5$).

Knowledge depth is defined as inventors’ expertise in a patent’s focal technological domain. We first identify all inventors for each patent, by coding inventor information from PatentsView, which employs probabilistic methodologies to disambiguate inventor records in a relational database (Whittington, 2018). Knowledge depth is then measured as the total number of patents in the *same technology class* as the focal patent that those inventors have developed in the past (Wang et al., 2017).

Knowledge breadth refer to the breadth of knowledge that inventors bring to the innovation effort. In line with prior studies (Fleming et al., 2007; Boh et al., 2014), we measure it as the number of unique technology classes in which the focal inventors have developed any patents prior to the focal patent.⁷

3.3 Control variables

We control for a set of factors that are commonly included in prior studies, which may otherwise influence the results (e.g., Nemet and Johnson, 2012; Gruber et al., 2013; Kaplan and Vakili, 2015; Wang et al., 2017). First, we calculate *team size*, the number of co-inventors in a patent; *inventors’ experience*, the total number of patents granted to inventors prior to the focal patent; and *invention tenure*, the average number of years between the first approved patent of each team member and the focal patent. At the patent level, we include the *number of subclass* assignments

⁷ To be precise, Fleming et al. (2007: 454) use the number of *subclasses*, rather than the number of *main classes*. We opt for the latter measure because the number of subclasses may not reflect the breadth of experience if they belong to the same main class.

for a patent. We also control for the *number of claims*. Patent claims define the scope and complexity of an invention for which patent protection is requested (Gruber *et al.*, 2013). To build this variable, we simply count the number of claims listed in each patent document. We also control for the *number of backward citations* of each patent that reflects the number of prior arts it draws upon. Finally, the *contrast* of technological domains could plausibly be related to several important factors in prior work. In order to rule out their potential confounding effects, we control for *technological distance* measured as the average of the distance between each pair of backward citations of a patent (Kaplan and Vakili, 2015), *technological diversity* measured as a Herfindahl index of the concentration of classes in the prior arts cited by a patent (Hall *et al.*, 2001), and *component familiarity* measured as the degree to which each of the subclasses in a focal patent have been recently and frequently used (Fleming, 2001).

4. Estimation and results

4.1 Estimation approach

We employ a fixed-effects estimation approach that helps account for heterogeneity across technology classes. While we focus on examining the role of class contrast, technology classes differ along many other dimensions that may confound our estimation. To address this concern, we use linear models with fixed-effects that perform particularly well with a large dataset and high-dimensional heterogeneity (*reghdfe* in Stata, (Correia, 2016)). Specially, by incorporating class $\times$ year fixed-effects, our models fully absorb technology class heterogeneity within each year and rule out any other class-level factors that may shape the results (e.g., class size and technological fertility). Because of the fixed-effects, only variables that vary within each class $\times$ year cell can be identified. This means that the main effect of domain contrast cannot be separately estimated because it does not vary within class $\times$ year cells. However, its interaction effects with other variables (e.g., domain spanning) can be estimated as they vary within class $\times$ year cells (see a similar approach by Samila and Sorenson (2011) and Bowers (2015)). To account for nonindependence, all standard errors are clustered at the class $\times$ year level.

We prefer the linear approach over generalized negative binomial models (Fleming and Sorenson, 2001) for several reasons. First, the linear models provide more efficient fixed-effects estimation that accounts for class heterogeneity, whereas the generalized approach does not support fixed-effects. Second, all of our hypotheses are formulated about interaction effects. While testing interaction effects is straightforward in linear models, estimating and interpreting interaction terms in limited dependent variable models can be much more complex (Wiersema and Bowen, 2009). Finally, the linear approach is a consistent estimator even with count data (Acs *et al.*, 2002; Ganco, 2017).

4.2 Results

Table 1 shows descriptive statistics and correlations. Estimation results are reported in Table 2. Models 1A and 1B include only control variables. In Models 2A and 2B, we test Hypothesis 1 by adding the interaction term between *contrast of tech domain* and *domain spanning (dummy)*. It has both a positive effect on the mean of forward citations ($\beta = 0.130$; $SE = 0.027$) and a positive effect on its variance ($\beta = 0.025$; $SE = 0.011$). Figure 2A depicts the marginal effects of *contrast of tech domain* and *domain spanning* on impact. When domain contrast is low (i.e., fuzzy domains), the effect of domain spanning is significantly negative. The effect turns significantly positive as domain contrast approaches its mean value. This suggests that while spanning fuzzy domains undermines inventive impact, spanning crisp domains helps enhance the impact of innovation. Figure 2B shows their marginal effects on uncertainty. When contrast is low, spanning domains does not bring much uncertainty. However, as contrast increases and approaches its mean value, the effect of domain spanning on uncertainty turns to become significantly positive. This suggests that spanning crisp boundaries leads to greater inventive uncertainty.

In Models 3A and 3B, we use a continuous measure of *domain spanning (ratio)* and find similar evidence as the interaction term between *contrast of tech domain* and *domain spanning (ratio)* has a positive effect on both the mean ($\beta = 0.189$; $SE = 0.050$) and the variance of

Table 1. Descriptive statistics

Variables	Mean	SD	Min.	Max.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 Forward citations	13.41	44.03	0.00	3471	1.00														
2 Contrast of domain	0.65	0.13	0.00	0.94	0.01	1.00													
3 Domain spanning (dummy)	0.39	0.49	0.00	1.00	0.01	-0.25	1.00												
4 Domain spanning (ratio)	0.20	0.27	0.00	0.98	0.01	-0.27	0.91	1.00											
5 Knowledge depth (ln)	1.82	1.65	0.00	11.04	0.04	0.14	-0.08	-0.09	1.00										
6 Knowledge breadth (ln)	1.87	0.84	0.00	5.12	0.00	0.02	0.02	0.02	0.47	1.00									
7 Inventors' experience (ln)	3.38	1.74	0.69	11.80	0.02	0.04	0.01	0.02	0.69	0.86	1.00								
8 Team size (ln)	0.83	0.62	0.00	4.33	0.04	-0.08	0.04	0.05	0.25	0.36	0.35	1.00							
9 Team tenure (ln)	2.06	0.70	0.07	3.56	-0.02	0.01	-0.01	-0.01	0.32	0.60	0.53	-0.09	1.00						
10 Backward citations (ln)	2.24	1.01	0.00	7.34	0.18	0.01	0.02	0.02	0.10	0.02	0.06	0.02	0.05	1.00					
11 Number of subclasses (ln)	1.34	0.54	0.69	4.60	0.01	-0.12	0.56	0.53	0.01	0.02	0.02	0.06	0.00	0.04	1.00				
12 Number of claims (ln)	2.73	0.69	0.69	6.50	0.11	-0.01	0.03	0.03	0.03	-0.03	0.00	0.04	-0.05	0.19	0.05	1.00			
13 Technology diversity	0.54	0.34	0.00	1.00	0.05	-0.10	0.16	0.16	-0.08	0.08	0.05	0.01	0.01	0.37	0.08	0.08	1.00		
14 Component familiarity	4.40	1.25	0.00	8.70	0.02	-0.10	0.15	0.13	0.13	0.04	0.10	0.07	-0.05	-0.09	0.14	0.03	0.01	1.00	
15 Technology distance	0.47	0.32	0.00	1.00	0.02	-0.22	0.24	0.26	-0.18	0.07	0.05	0.06	0.00	0.00	0.11	0.02	0.40	0.04	1.00

Table 2. Boundary spanning and the contrast of tech classes

Variables	Mean Model 1A	Variance Model 1B	Mean Model 2A	Variance Model 2B	Mean Model 3A	Variance Model 3B
Contrast of tech domain × domain spanning (ratio)					0.189 ^{***} (0.050)	0.049 [*] (0.020)
Contrast of tech domain × domain spanning (dummy)			0.130 ^{***} (0.027)	0.025 [*] (0.011)		
Domain spanning (ratio)					-0.086 ^{**} (0.032)	-0.022 [†] (0.013)
Domain spanning (dummy)	0.015 ^{***} (0.004)	0.004 [*] (0.002)	-0.068 ^{***} (0.018)	-0.012 [†] (0.007)		
Knowledge depth (ln)	0.012 ^{***} (0.002)	0.000 (0.001)	0.012 ^{***} (0.002)	0.000 (0.001)	0.012 ^{***} (0.002)	0.001 (0.001)
Knowledge breadth (ln)	-0.047 ^{***} (0.005)	-0.020 ^{***} (0.002)	-0.047 ^{***} (0.005)	-0.020 ^{***} (0.002)	-0.047 ^{***} (0.005)	-0.020 ^{***} (0.002)
Inventors' experience (ln)	0.026 ^{***} (0.003)	0.017 ^{***} (0.001)	0.026 ^{***} (0.003)	0.017 ^{***} (0.001)	0.026 ^{***} (0.003)	0.017 ^{***} (0.001)
Team size (ln)	0.125 ^{***} (0.003)	0.018 ^{***} (0.001)	0.125 ^{***} (0.003)	0.018 ^{***} (0.001)	0.125 ^{***} (0.003)	0.018 ^{***} (0.001)
Team tenure (ln)	-0.014 ^{***} (0.003)	-0.014 ^{***} (0.001)	-0.014 ^{***} (0.003)	-0.014 ^{***} (0.001)	-0.014 ^{***} (0.003)	-0.014 ^{***} (0.001)
Backward citations (ln)	0.296 ^{***} (0.003)	0.045 ^{***} (0.001)	0.296 ^{***} (0.003)	0.045 ^{***} (0.001)	0.296 ^{***} (0.003)	0.045 ^{***} (0.001)
Number of subclasses (ln)	0.056 ^{***} (0.004)	0.009 ^{***} (0.001)	0.057 ^{***} (0.004)	0.009 ^{***} (0.001)	0.056 ^{***} (0.004)	0.009 ^{***} (0.001)
Number of claims (ln)	0.274 ^{***} (0.003)	0.047 ^{***} (0.001)	0.274 ^{***} (0.003)	0.047 ^{***} (0.001)	0.274 ^{***} (0.003)	0.047 ^{***} (0.001)
Technology diversity	-0.086 ^{***} (0.006)	0.019 ^{***} (0.002)	-0.086 ^{***} (0.006)	0.019 ^{***} (0.002)	-0.086 ^{***} (0.006)	0.019 ^{***} (0.002)
Component familiarity	0.038 ^{***} (0.003)	0.011 ^{***} (0.001)	0.038 ^{***} (0.003)	0.011 ^{***} (0.001)	0.038 ^{***} (0.003)	0.011 ^{***} (0.001)
Technology distance	0.121 ^{***} (0.009)	0.010 ^{***} (0.003)	0.120 ^{***} (0.009)	0.010 ^{***} (0.003)	0.118 ^{***} (0.009)	0.009 ^{***} (0.003)
Constant	-0.232 ^{***} (0.018)	0.428 ^{***} (0.007)	-0.232 ^{***} (0.018)	0.428 ^{***} (0.007)	-0.230 ^{***} (0.018)	0.429 ^{***} (0.007)
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,166,436	1,166,436	1,166,436	1,166,436	1,166,436	1,166,436

Robust standard errors clustered at the class × year level in parentheses.

*** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, and † $P < 0.10$.

forward citations ($\beta = 0.049$; $SE = 0.020$). Altogether, these results support both Hypothesis 1A— inventions that span crisp boundaries are on average more impactful than inventions that span fuzzy boundaries—and Hypothesis 1B— inventions that span crisp boundaries are associated with higher uncertainty than inventions that span fuzzy boundaries.

Table 3 presents the results for Hypotheses 2 and 3 that explore whether knowledge depth and breadth are helpful for spanning crisp boundaries. In this set of estimations, we analyze all boundary-spanning patents in the sample (in extensional analyses as mentioned later, we also report results for patents without spanning domains). In Models 2A and 2B, we add the interaction term between *knowledge depth* and *contrast of tech domain* to test Hypothesis 2. In Model 2A, the interaction has a significant and positive effect on mean citations ($\beta = 0.038$; $SE = 0.015$). Specifically, one SD increase of knowledge depth helps enhance the impact of patents spanning crisp boundaries by about 6.5% ($= \exp(0.038 \times 1.65)$). This is consistent with our Hypothesis 2A that knowledge depth enhances the impact of inventions that span crisp boundaries. In Model

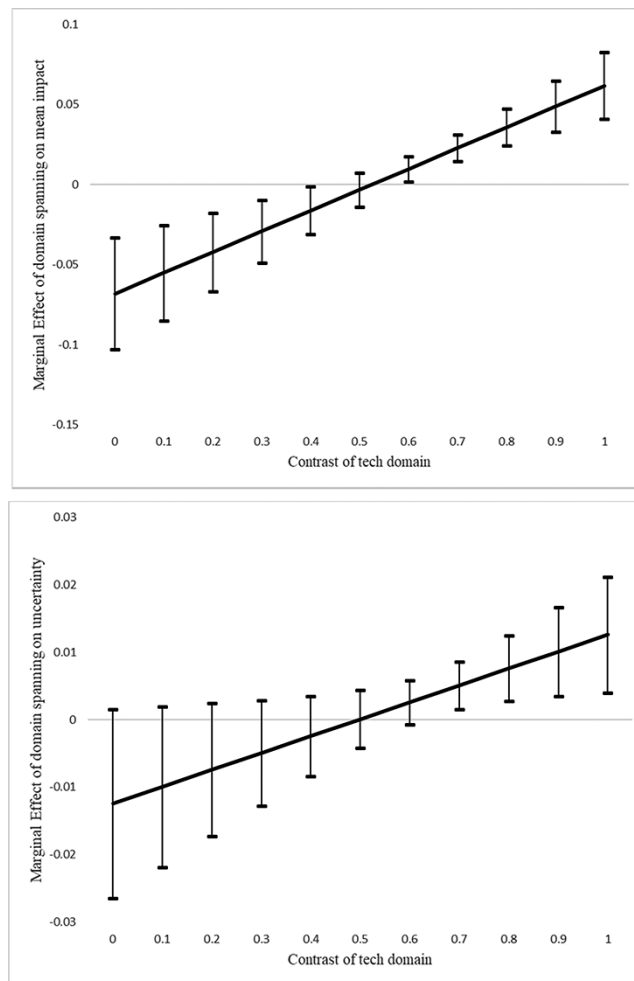


Figure 2. (A) Domain spanning and contrast on mean impact and (B) domain spanning and contrast on uncertainty

2B, the interaction term has a negative but nonsignificant effect on uncertainty ($\beta = -0.001$; $SE = 0.007$). This suggests that knowledge depth does not significantly influence the uncertainty of inventions that span crisp boundaries. Thus, our Hypothesis 2B is not supported.

In Models 3A and 3B, we add the interaction term between *knowledge breadth* and *contrast of tech domain* to test Hypothesis 3. In Model 3A, its effect on mean citations is positive and significant ($\beta = 0.091$; $SE = 0.023$). It suggests that one SD increase of knowledge breadth helps enhance the impact of patents spanning crisp boundaries by about 7.9% ($=\exp(0.091 \times 0.84)$). This is opposite to our Hypothesis 3A that knowledge breadth decreases the impact of inventions that span crisper boundaries. In Model 3B, the interaction term has a positive effect on variance ($\beta = 0.023$; $SE = 0.009$). As knowledge breadth increases, the effect of contrast on boundary-spanning inventions' uncertainty increases. One SD increase of knowledge breadth leads the recombinant uncertainty of patents spanning crisp boundaries to increase about 1.9% ($=\exp(0.023 \times 0.84)$). Our Hypothesis 3B is hence supported that knowledge breadth increases the recombinant uncertainty of inventions that span crisp boundaries.

In sum, the results suggest that knowledge depth and breadth are both important for inventors to span crisp boundaries. However, knowledge breadth also escalates the uncertainty associated with spanning crisp boundaries, whereas knowledge depth does not show significant influence on the uncertainty.

Table 3. Knowledge depth and breadth on domain spanning

Variables	Mean Model 1A	Variance Model 1B	Mean Model 2A	Variance Model 2B	Mean Model 3A	Variance Model 3B	Mean Model 4A	Variance Model 4B
Contrast of tech domain × knowledge depth (ln)	0.038 [*] (0.015)	-0.001 (0.007)					0.021 (0.018)	-0.009 (0.008)
Contrast of tech domain × knowledge breadth (ln)			0.091 ^{***} (0.023)			0.023 [*] (0.009)	0.073 [*] (0.028)	0.030 ^{***} (0.012)
Knowledge depth (ln)	0.009 ^{***} (0.002)	-0.000 (0.001)	-0.015 (0.010)	0.000 (0.004)	0.008 ^{***} (0.002)	-0.000 (0.001)	-0.004 (0.012)	0.005 (0.005)
Knowledge breadth (ln)	-0.038 ^{***} (0.008)	-0.021 ^{***} (0.003)	-0.038 ^{***} (0.008)	-0.021 ^{***} (0.003)	-0.093 ^{***} (0.016)	-0.035 ^{***} (0.006)	-0.082 ^{***} (0.020)	-0.040 ^{***} (0.008)
Inventors' experience (ln)	0.022 ^{***} (0.004)	0.017 ^{***} (0.001)	0.022 ^{***} (0.004)	0.016 ^{***} (0.001)	0.022 ^{***} (0.004)	0.016 ^{***} (0.001)	0.022 ^{***} (0.004)	0.016 ^{***} (0.001)
Team size (ln)	0.128 ^{***} (0.004)	0.016 ^{***} (0.002)	0.129 ^{***} (0.004)	0.016 ^{***} (0.002)	0.128 ^{***} (0.004)	0.016 ^{***} (0.002)	0.128 ^{***} (0.004)	0.016 ^{***} (0.002)
Team tenure (ln)	-0.015 ^{***} (0.005)	-0.012 ^{***} (0.002)	-0.015 ^{***} (0.005)	-0.012 ^{***} (0.002)	-0.015 ^{***} (0.005)	-0.012 ^{***} (0.002)	-0.015 ^{***} (0.005)	-0.012 ^{***} (0.002)
Backward citations (ln)	0.309 ^{***} (0.004)	0.041 ^{***} (0.002)	0.309 ^{***} (0.004)	0.041 ^{***} (0.002)	0.309 ^{***} (0.004)	0.041 ^{***} (0.002)	0.309 ^{***} (0.004)	0.042 ^{***} (0.002)
Number of subclasses (ln)	0.070 ^{***} (0.005)	0.010 ^{***} (0.002)	0.071 ^{***} (0.005)	0.011 ^{***} (0.002)	0.071 ^{***} (0.005)	0.011 ^{***} (0.002)	0.071 ^{***} (0.005)	0.011 ^{***} (0.002)
Number of claims (ln)	0.281 ^{***} (0.004)	0.042 ^{***} (0.002)	0.281 ^{***} (0.004)	0.042 ^{***} (0.002)	0.281 ^{***} (0.004)	0.042 ^{***} (0.002)	0.281 ^{***} (0.004)	0.042 ^{***} (0.002)
Technology diversity	-0.087 ^{***} (0.008)	0.022 ^{***} (0.004)	-0.087 ^{***} (0.008)	0.022 ^{***} (0.004)	-0.087 ^{***} (0.008)	0.022 ^{***} (0.004)	-0.087 ^{***} (0.008)	0.022 ^{***} (0.004)
Component familiarity	0.064 ^{***} (0.005)	0.022 ^{***} (0.001)	0.064 ^{***} (0.005)	0.022 ^{***} (0.001)	0.064 ^{***} (0.005)	0.022 ^{***} (0.001)	0.064 ^{***} (0.005)	0.022 ^{***} (0.001)
Technology distance	0.090 ^{***} (0.012)	0.004 (0.004)	0.090 ^{***} (0.012)	0.004 (0.004)	0.089 ^{***} (0.012)	0.004 (0.004)	0.089 ^{***} (0.012)	0.004 (0.004)
Constant	-0.398 ^{***} (0.028)	0.401 ^{***} (0.010)	-0.398 ^{***} (0.028)	0.400 ^{***} (0.010)	-0.396 ^{***} (0.028)	0.400 ^{***} (0.010)	-0.396 ^{***} (0.028)	0.401 ^{***} (0.010)
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	459,172	459,172	459,172	459,172	459,172	459,172	459,172	459,172

Robust standard errors clustered at the class × year level in parentheses.

*** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, and † $P < 0.10$.

4.3 Robustness checks and extensions

In addition to the main analyses, we have conducted a number of additional analyses to check the robustness of results, explore further the data, and deepen our understandings.

4.3.1 Alternative measurement

When counting the number of forward citations, we have followed prior studies to consider all citations until the end year in the database (Yayavaram and Ahuja, 2008; Carnabuci *et al.*, 2015). This may lead to a potential concern that earlier patents have longer time periods to receive citations. Although our estimation incorporates year fixed-effects to address this concern, we adopt alternative approach that calculates each patent’s first 5-year forward citations. The results are reported in Panel A of Table 4, which are largely consistent in terms of both statistical significance and economic relevance. Moreover, in analyses not tabulated here, we also exclude self-citations when counting forward citations (Higham *et al.*, 2021) (which has a correlation of 0.799 with our main measure) and find largely similar results.

Relatedly, in our main estimation, we use inventors’ experience of their entire careers to measure knowledge breadth and depth. To examine the sensitivity of results, we recalculate the two variables based on inventors’ *recent* experience (i.e., 5 years prior to the focal patent) and report the results in Panel B of Table 4, which show quite consistent results.

4.3.2 Domain contrast or boundary spanning

We further decompose the process of spanning crisp boundaries, which consist of two parts—boundary spanning and domain contrast. While we find evidence for the roles of knowledge depth and breadth in spanning crisp boundaries, it is unclear whether their effects are mainly related to the process of boundary spanning or related to the contrast of domains. To dig into this, we conduct two extensional analyses. First, we simply repeat our analysis of knowledge depth and breadth in the sample of non-domain-spanning patents as well as in the full sample. The results are reported in Panel C of Table 4, which is largely similar to our findings in Table 3 from the sample of domain-spanning patents. This hints that while knowledge depth and breadth do affect inventions that span crisp domains, their effects are less dependent on whether inventions are domain spanning but more related to the degree of domain contrast.

Indeed, this is corroborated in our second analysis in Panel D of Table 4 that utilizes triple interactions from the full sample. While the interaction effects of *knowledge depth/breadth* with *contrast of tech domain* are significant, their interactions with *domain spanning* are nonsignificant and the two three-way interactions are also marginal. For instance, in Model 1A, the interaction between depth and contrast is significantly positive ($\beta = 0.035$; $SE = 0.013$), suggesting that knowledge depth is helpful to enhance the impact of patents in crisp domains; the interaction between depth and domain spanning is nonsignificant ($\beta = -0.011$; $SE = 0.009$), suggesting that knowledge depth does not significantly shape the impact of domain-spanning patents; and the triple interaction is nonsignificant ($\beta = 0.013$; $SE = 0.0014$), suggesting that domain spanning does not further strengthen the value of knowledge depth for innovating in crisp domains. Taken together, this suggests that the importance of inventors’ knowledge depth and breadth is mostly contingent on the contrast of a technological domain, with less regard to whether they span the domain or not.

4.3.3 Breakthroughs versus failures

While we have analyzed recombinant uncertainty by predicting variance, we also want to further decompose it (Singh and Fleming, 2010). That is because greater variance may indicate a higher chance of inventive failures, a larger likelihood of technological breakthroughs, and/or a wider variance in the middle of the distribution. Following prior studies (Singh and Fleming, 2010), we define a patent as a failure if it receives no forward citations ($FL = 1$, and 0 otherwise), and a patent as a breakthrough if it is in the top 5% in terms of forward citations, compared with patents from the same class in the same year ($BKT = 1$, and 0 otherwise). We then estimate the likelihood of each of them using fixed-effects linear probability models (Bowers, 2015) and report the results in Table 5.

Table 4. Robustness checks

Panel A. Forward citations in 5 years					
Variables	Mean Model 1A	Variance Model 1B	Mean Model 2A	Variance Model 2B	Mean Model 3A Variance Model 3B
Contrast of tech domain × domain spanning (dummy)	0.098 ^{***} (0.021)	0.036 ^{***} (0.009)			
Contrast of tech domain × knowledge depth (ln)			0.031 ^{**} (0.012)	−0.004 (0.005)	
Contrast of tech domain × knowledge breadth (ln)					0.080 ^{***} (0.018)
Other controls	Yes	Yes	Yes	Yes	Yes
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes
Observations	1,166,436	1,166,436	424,824	424,824	424,824
Panel B. Knowledge depth and breadth in 5 years					
Variables	Mean Model 1A	Variance Model 1B	Mean Model 2A	Variance Model 2B	Mean Model 3A Variance Model 3B
Contrast of tech domain × knowledge depth (ln)	0.052 [*] (0.023)	−0.003 (0.010)			0.032 (0.027)
Contrast of tech domain × knowledge breadth (ln)			0.094 ^{***} (0.023)	0.017 [†] (0.010)	0.074 ^{**} (0.028)
Knowledge depth (ln)	−0.024 (0.015)	0.003 (0.006)	0.008 [*] (0.003)	0.002 (0.001)	−0.012 (0.017)
Knowledge breadth (ln)	−0.008 (0.007)	−0.007 ^{**} (0.002)	−0.065 ^{***} (0.016)	−0.018 ^{**} (0.007)	−0.053 ^{***} (0.019)
Other controls	Yes	Yes	Yes	Yes	Yes
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes
Observations	1,166,436	1,166,436	424,824	424,824	424,824

(continued)

Table 4. (Continued)

Variables	Panel C. Knowledge depth and breadth in non-domain spanning							
	Non-domain spanning				Full sample			
	Model 1A	Model 1B	Model 2A	Model 2B	Model 3A	Model 3A	Model 4A	Model 4B
Contrast of tech domain × knowledge depth (ln)	0.041 ^{***} (0.013)	0.013 [*] (0.005)			0.041 ^{***} (0.012)	0.009 [†] (0.005)		
Contrast of tech domain × knowledge breadth (ln)			0.112 ^{***} (0.020)	0.028 ^{***} (0.008)			0.098 ^{***} (0.017)	0.028 ^{***} (0.007)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	706,976	706,976	706,976	706,976	1,166,436	1,166,436	1,166,436	1,166,436
Panel D. Triple interactions								
Variables	Model 1A	Model 1B	Model 2A	Model 2B				
Contrast of tech domain × domain spanning (dummy) × knowledge depth (ln)	0.013 (0.014)	-0.015 [*] (0.006)						
Contrast of tech domain × domain spanning (dummy) × knowledge breadth (ln)			-0.033 (0.025)	-0.011 (0.011)				
Contrast of tech domain × knowledge depth (ln)	0.035 ^{***} (0.013)	0.013 [*] (0.005)						
Contrast of tech domain × knowledge breadth (ln)			0.119 ^{***} (0.020)	0.032 ^{***} (0.008)				
Domain spanning (dummy) × knowledge depth (ln)	-0.011 (0.009)	0.007 [†] (0.004)						
Domain spanning (dummy) × knowledge breadth (ln)			0.031 [†] (0.017)	0.008 (0.007)				
Contrast of tech domain × domain spanning (dummy)	0.114 ^{***} (0.034)	0.053 ^{***} (0.016)	0.187 ^{***} (0.057)	0.042 (0.026)				
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,166,436	1,166,436	1,166,436	1,166,436				

Robust standard errors clustered at the class × year level in parentheses.
^{***} $P < 0.001$, ^{**} $P < 0.01$, ^{*} $P < 0.05$, and [†] $P < 0.10$.

Table 5. Inventive breakthrough and failure

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	Full sample		FL	BKT	Domain spanning	
	FL	BKT			FL	BKT
Contrast of tech domain × domain spanning (dummy)	−0.027** (0.009)	0.010* (0.004)				
Contrast of tech domain × knowledge depth (ln)			−0.013* (0.005)	0.001 (0.002)		
Contrast of tech domain × knowledge breadth (ln)					−0.017* (0.008)	0.004 (0.003)
Domain spanning (dummy)	0.012* (0.006)	−0.004† (0.003)				
Knowledge depth (ln)	−0.004*** (0.001)	−0.000 (0.000)	0.005 (0.003)	−0.000 (0.001)	−0.003*** (0.001)	0.000 (0.000)
Knowledge breadth (ln)	0.006*** (0.002)	−0.006*** (0.001)	0.001 (0.002)	−0.007*** (0.001)	0.011* (0.005)	−0.009*** (0.002)
Inventors' experience (ln)	−0.000 (0.001)	0.005*** (0.000)	0.002 (0.001)	0.005*** (0.001)	0.002 (0.001)	0.005*** (0.001)
Team size (ln)	−0.031*** (0.001)	0.010*** (0.001)	−0.031*** (0.001)	0.010*** (0.001)	−0.031*** (0.001)	0.010*** (0.001)
Team tenure (ln)	−0.001 (0.001)	−0.003*** (0.000)	−0.001 (0.001)	−0.004*** (0.001)	−0.001 (0.001)	−0.004*** (0.001)
Backward citations (ln)	−0.060*** (0.001)	0.029*** (0.001)	−0.061*** (0.001)	0.030*** (0.001)	−0.061*** (0.001)	0.030*** (0.001)
Number of subclasses (ln)	−0.017*** (0.001)	0.001† (0.001)	−0.021*** (0.002)	0.002† (0.001)	−0.021*** (0.002)	0.002† (0.001)
Number of claims (ln)	−0.066*** (0.001)	0.018*** (0.001)	−0.065*** (0.001)	0.020*** (0.001)	−0.065*** (0.001)	0.020*** (0.001)
Technology diversity	0.006*** (0.002)	−0.013*** (0.001)	0.006* (0.003)	−0.015*** (0.001)	0.006* (0.003)	−0.015*** (0.001)
Component familiarity	−0.007*** (0.001)	0.004*** (0.000)	−0.010*** (0.001)	0.007*** (0.001)	−0.010*** (0.001)	0.007*** (0.001)
Technology distance	−0.006* (0.003)	0.018*** (0.001)	−0.001 (0.003)	0.015*** (0.001)	−0.001 (0.003)	0.015*** (0.001)
Constant	0.660***	−0.090***	0.669***	−0.114***	0.668***	−0.114***
Tech class × year fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,166,436	1,166,436	459,172	459,172	459,172	459,172

Robust standard errors clustered at the class × year level in parentheses.

*** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, and † $P < 0.10$.

FT, failure; BKT, breakthrough.

In Models 1 and 2, the interaction between contrast and domain spanning is negative and positive, respectively. This suggests that spanning crisp domains is useful to both reduce the likelihood of inventive failure and enhance the possibility of technological breakthroughs. As such, the high uncertainty associated with spanning crisp boundaries in our main estimation seems to be located at the upper tail of the distribution. However, this does not ascertain that spanning crisp boundaries is beneficial since we cannot see variance in the middle of the distribution. In Models 3–6, the interactions have a significantly negative effect on inventive failure but no significant effect on technological breakthroughs. This implies that while both knowledge depth and breadth are important to reduce the failure likelihood of patents that span crisp domains, they do not show significant value to increase the chances of achieving breakthroughs.

5. Discussion

This study draws attention to the boundaries of technological domains in the boundary-spanning innovation process. More specifically, we focus on the implications of spanning *crisp* versus *fuzzy* boundaries for the impact and uncertainty of inventions. Examining US patents, we show that spanning *crisp* boundaries is more likely to generate impactful innovations but at the same time leads to higher inventive uncertainty. Analyzing inventors’ knowledge expertise, we find that while both knowledge depth and breadth are useful to enhance the impact of technologies that span crisp boundaries, knowledge breadth is also found to escalate the associated uncertainty. Our study provides several implications for innovation theories and practices.

5.1 Crisp boundaries and technology islands

Prior literature has long underscored the importance of boundary spanning for innovation (Schumpeter, 1939; Fleming, 2001; Nemet and Johnson, 2012; Kaplan and Vakili, 2015) but paid less attention to the heterogeneity of those boundaries. Building upon recent studies on categorical contrast (Carnabuci *et al.*, 2015; Kovács *et al.*, 2021), we have distinguished between crisp and fuzzy domain boundaries. Conceptualizing crisp domains as “islands” in the technological landscape, this paper employs the knowledge recombination perspective (Xiao *et al.*, 2022) to uncover the important outcomes of innovation on islands. Our results show that when inventions span crisp boundaries to go beyond focal islands, their impact is significantly enhanced. As such, *boundary-spanning* recombination is particularly important for inventors (or firms) in technological islands in order to generate impactful inventions. However, our findings show also that inventive uncertainty is substantially escalated when inventions span the crisp boundaries of islands. This supports our core claim on the recombinative nature in crisp domains: technologies on islands are so distinctive from external technologies that combining them together for reproduction is substantially unpredictable. In other words, spanning crisp boundaries is a double-edged sword: while the isolation nature of technology islands may enhance the impact of boundary-spanning innovation, it also escalates difficulty and uncertainty in the recombination (or reproduction) process. As such, inventors or firms may have to weigh the potential gains from spanning crisp boundaries against the associated uncertainty.

Our findings also emphasize the importance of examining domain boundaries in the boundary-spanning innovation process. Prior research has extensively investigated the effect of boundary spanning on novelty, usefulness, and approval (Fleming, 2001; Kaplan and Vakili, 2015; Lo and Kennedy, 2015). Our study emphasizes further that innovation outcomes are not only related to the *degree* of boundary spanning, but are also largely dependent on the features of boundaries being spanned. While this paper focuses on the contrast of domain boundaries (i.e., crisp versus fuzzy), future studies may extend attention to other important dimensions that characterize domain-specific knowledge boundaries.

Although this paper builds upon recent development on the contrast of technological domains (Kovács *et al.*, 2021), it also adds to this line of inquiry in at least three ways. First, while previous studies focus on the implications of belonging to crisp versus fuzzy classes (i.e., the main effect of contrast) (Carnabuci *et al.*, 2015; Kovács *et al.*, 2021), our research moves on to explore how boundary spanning alters the importance of domain contrast in technological innovation. Second, whereas earlier work investigates how domain contrast influences the mean performance of innovation, we also draw attention to the variance or uncertainty of innovation outputs (Taylor and Greve, 2006). Finally, as elaborated later, we also shift attention from testing the performance implications of domain contrast to exploring what agents are better able to innovate in crisp or fuzzy technological domains. Such exploration is particularly important for inventors (or firms) to understand what skills to accumulate in different types of technological domains.

5.2 Knowledge expertise and boundary spanning

Recognizing both gains and uncertainty associated with spanning crisp boundaries, we have explored how inventors may better reap the benefits with less uncertainty, by examining the roles of knowledge depth and breadth. Whereas prior research has established their generic roles in the innovation process (Katila and Ahuja, 2002; Carnabuci and Operti, 2013; Boh *et al.*, 2014;

Melero and Palomer, 2015), this study focuses on their specific roles in spanning crisp boundaries. Our findings highlight that deep knowledge in focal domains can help inventors enhance the impact of inventions that span crisp boundaries, albeit with no significant effect on the associated uncertainty. As such, to innovate beyond the boundaries of technological islands, it is important for inventors to gain deeper insights about technologies on those islands.

The role of knowledge breadth is a bit complex. It is well established that knowledge breadth contributes to boundary spanning because it increases the possibility of generating novel combinations (Fleming *et al.*, 2007; Carnabuci and Operti, 2013; Gruber *et al.*, 2013). Our distinction between crisp and fuzzy boundaries helps extend this research. We find that knowledge breadth does not contribute *equally* to any type of boundary-spanning innovations. Rather, broad knowledge scope is found to both enhance the value of and escalate the uncertainty of innovations that span crisp boundaries, as compared to innovations that span fuzzy boundaries. In fact, the findings that knowledge breadth enhances the impact of innovation spanning are in contrast to our *ex-ante* hypothesis. There are at least two possible explanations for it. First, broad knowledge scope may be more useful for innovations in crisp domains, as inventors with a narrow scope are likely to suffer from intellectual myopia or competence traps (Levinthal and March, 1993). Second, inventors with broad knowledge may find it easier to generate novel innovations in crisp domains because it causes comparatively less cognitive overload in such domains than in fuzzy domains.

Overall, those findings also add to debate on the role of generalists versus specialists in innovation (i.e., knowledge breadth versus depth) (Katila and Ahuja, 2002; Jones, 2009; Melero and Palomer, 2015). While some scholars emphasize the importance of knowledge specialization (Jones, 2009), others argue that generalist inventors may perform better because of their advantage in combining knowledge (Melero and Palomer, 2015). Adding to this research, our study suggests that there may be no absolute difference between specialists and generalists in creating impactful innovations. Rather, their value is contingent on how crisp the domain boundaries of innovations are.

5.3 Future studies

Several important questions remain to be answered. First, while emphasizing the contrast of primary technological domains, we have to downplay the heterogeneity of secondary domains that are used for recombination. For each boundary-spanning innovation, there is one primary domain that defines its core function and one or several secondary domains that provide complementary technologies (Fleming, 2001; Carnabuci *et al.*, 2015). In the current framework, we distinguish between crisp and fuzzy primary domains but treat secondary domains as homogeneous. It would be interesting to extend our analysis and distinguish between crisp and fuzzy secondary domains. More specifically, future studies can focus on the composite structure in inventions, by comparing different primary-secondary domain combinations (i.e., crisp-crisp, crisp-fuzzy, fuzzy-crisp, and fuzzy-fuzzy combinations). In other words, when voyaging beyond a technology island, is it better for inventors to reach out to other isolated islands or densely connected archipelagoes?

Second, in analyzing the roles of knowledge depth and breadth, we have an implicit assumption that the composition of inventors for an invention is exogenous (e.g., Gruber *et al.*, 2013). However, some unobserved factors may influence both what types of inventors tend to work together and the usefulness and uncertainty of inventions, leading to concerns about potential endogeneity issues (Melero and Palomer, 2015). While we have controlled for a large number of factors commonly used in prior studies, it is important to acknowledge that the list may not be complete. Future research is hence welcome to improve our estimation approach.

Third, another interesting direction is to explore the exact mechanisms for our findings. While our study focuses on the effect of spanning crisp boundaries on innovation outcomes, we have not teased out the mechanisms through which it takes place. Future studies, for instance, may investigate whether spanning crisp boundaries leads to greater novelty (Kaplan and Vakili, 2015; Aharonson and Schilling, 2016), atypicality (Uzzi *et al.*, 2013), and other mediating factors in order to better understand the effect paths. Moreover, future work is needed to distinguish the producer-side effect of domain contrast from the audience-side effect in the innovation process.

(Kovács *et al.*, 2021). Put differently, while we establish the importance of contrast in boundary-spanning innovation, it is important to examine whether contrast changes production quality (Carnabuci *et al.*, 2015) or influences the audience-side cognition (Kovács *et al.*, 2021), or both.

Finally, there are several general issues related to patent data. Given data availability, we are only able to analyze granted patents but cannot observe the dynamics in patent examination processes. Based on prior work (Ferguson and Carnabuci, 2017), one may expect similar effects of contrast on patent approval. As a result, there is less variance among the granted patents than variance among all inventions. Our estimation may thus be considered more conservative since we are less likely to find support from only granted patents. Nonetheless, we encourage future work to explore patent examination processes using novel data. Moreover, assigning patents to technology classes is up to patent examiners’ judgment, which is to some extent subjective. Although we believe that the combination of the USPTO classification system with examiners’ expertise yields a higher degree of accuracy and reliability (Carnabuci and Bruggeman, 2009), future studies may test our framework in other contexts where innovations are more objectively assigned to domains. Furthermore, it is worth noting that the classification system is not static but dynamic with new classes being introduced and old classes being discontinued (Lerner and Seru, 2017). For the purpose of this study on technological recombination, we have used patents’ original classifications. The dynamics provide an interesting opportunity for future studies to examine the impact of system changes on recombinative innovation.

5.4 Conclusion

Despite the aforementioned limitations, our study draws attention to the heterogeneity of domain boundaries, distinguishes between crisp and fuzzy boundaries in the boundary-spanning innovation process, and examines the roles of knowledge depth and breadth for innovations spanning crisp versus fuzzy boundaries. Our results show that boundary spanning enhances the impact of innovation on technology islands (i.e., domains with crisp boundaries) but at the same time escalates inventive uncertainty. While both knowledge depth and breadth can help enhance the impact of spanning crisp boundaries, knowledge breadth may escalate the associated uncertainty.

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